

The elements with atomic numbers 89 to 103 are the actinides.

# Thorium

**90 Th** A relatively common element, thorium appears in many naturally occurring minerals. Special glass with thorium dioxide was once used to make high-efficiency camera lenses, but the thorium made them radioactive. Research into using thorium as a nuclear fuel is ongoing.



Brownish-red mineral

## Monazite

The chief ore of thorium, monazite also contains varying amounts of the elements cerium, lanthanum, gadolinium, and neodymium. Depending on the exact composition, monazite varies in color from yellow to brownish-red.

# Actinides

The elements in this series are fiercely radioactive metals. Most of them are either present only as decayed radioactive products or have to be artificially created. All the elements after uranium are called “transuranium” elements, and most of them are used mainly in research.

# Actinium

**89 Ac** A radioactive element present in Earth’s crust in tiny amounts, actinium was first isolated in 1902. It gives the name to the actinide series. Because it is so scarce, there aren’t many uses for it, but the isotope actinium-225 is being studied for use in some cancer treatments.



Yellow-green crystals under visible light

## Autunite

Actinium can be found in autunite, a mineral that contains uranium. The uranium in the rock undergoes natural radioactive decay, and in the process, it produces actinium atoms in small quantities. Autunite’s crystals glow bright green under ultraviolet light.

These vials for intravenous injections contain actinium, selenium, and radium.



## Radioactive cures

In the early 20th century, radiation was considered by some as a cure-all for many diseases, and intravenous injections for various illnesses contained radioactive elements, including actinium. As radioactivity became better understood, it was discovered that these treatments had terrible health consequences.